RAMCO INSTITUTE OF TECHNOLOGY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

COURSE OUTCOMES

REGULATION 2021

COURSE OUTCOMES – BATCH 2021-2025	
I Year	
Course Code & Title : HS3152 Professional English - I **Semester & Year of study : I & 2021-22**	
Course Outcomes	
CO1	Communicate clearly both in written and oral forms using appropriate vocabulary and comprehend written text to make inferences.
CO2	Speak persuasively in academic/work contexts and write biographical details and technical documents cohesively, coherently, and flawlessly using appropriate words.
CO3	Read, write and speak effectively in a variety of professional and social settings.
CO4	Comprehend descriptive, narrative, expository and interpretive texts and write using creative, critical, analytical, and evaluative methods.
CO5	Understand and respond to different spoken and written discourses/excerpts in different accents and write different genres of text adopting various writing strategies.
CO6	Detect paralinguistic cues such as postures, gestures, facial expressions, and eye contact.
Course Code & Title : MA3151 Matrices and Calculus **Semester & Year of study : I & 2021-22**	
Course Outcomes	
CO1	Use the matrix algebra methods for solving practical problems.
CO2	Apply differential calculus tools in solving various application problems.
CO3	Use differential calculus ideas on several variable functions.
CO4	Apply different methods of integration in solving practical problems.
CO5	Apply multiple integral ideas in solving areas, volumes and other practical problems.
CO6	Use MATLAB for finding eigenvalues of a Matrix, derivative and integration of functions.
Course Code & Title : PH3151 Engineering Physics **Semester & Year of study : I & 2021-22**	
Couse Index	**Course Outcomes**
CO1	Understand the importance of mechanics.

CO2	Express their knowledge in electromagnetic waves.
CO3	Demonstrate a strong foundational knowledge in oscillations, optics and lasers.
CO4	Understand the importance of quantum physics.
CO5	Comprehend and apply quantum mechanical principles towards the formation of energy bands.
CO6	Build a simulation models to calculate the energy values of particle in a 1D, 2D & 3D boxes using computer programming.

Course Code & Title : CY3151 Engineering Chemistry
Semester & Year of study : I & 2021-22

Course Outcomes	
CO1	Illustrate the importance of water quality parameters, water treatment methods and boiler troubles.
CO2	Explain the types of nanomaterials, synthesis and its applications.
CO3	Apply the use of phase rule in metallurgy, basics of composites & its applications.
CO4	Articulate the fuel types, synthesis and its combustion characteristics
CO5	Examine the working principle of alternate energy resources and energy storage devices.
CO6	Analyze the water quality parameters of water samples collected in and around the native area

Course Code & Title : GE3151 Problem Solving and Python Programming
Semester & Year of study : I & 2021-22

Course Outcomes	
CO1	Develop algorithmic solutions to simple computational problems.
CO2	Demonstrate simple Python programs.
CO3	Apply control structures, functions and string to write simple program for solving problems.
CO4	Implement compound data using Python lists, tuples, dictionaries etc.
CO5	Illustrate read and write data from/to files in Python programs.
CO6	Build an application by applying various Python concepts.

Course Code & Title : GE3171 Problem Solving and Python Programming Laboratory
Semester & Year of study : I & 2021-22

Course Outcomes	
CO1	Develop diagrammatic solutions to simple computational problems
CO2	Execute simple Python programs using simple statements, operators and expressions
CO3	Implement various computing strategies for Python-based solutions to real world problems.
CO4	Apply compound data using python lists, tuples, sets and dictionaries to solve real life problems
CO5	Implement read and write data from/to files in python programs

Course Code & Title : BS3171 Physics and Chemistry Laboratory
Semester & Year of study : I & 2021-22

Course Outcomes	
CO1	Understand the functioning of various physics laboratory equipment.
CO2	Use graphical models to analyze laboratory data.
CO3	Use mathematical models as a medium for quantitative reasoning and describing physical reality.
CO4	Access, process and analyze scientific information.
CO5	Solve problems individually and collaboratively.
CO6	Analyze the quality of the water samples like hardness, alkalinity, chloride and dissolved oxygen content.
CO7	Quantitatively examine the impurities present in solutions by electro-analytical techniques.
CO8	Practice the synthesis of nanoparticles.

Course Code & Title : GE3172 English Laboratory
Semester & Year of study : I & 2021-22

Course Outcomes	
CO1	Listen and comprehend complex academic texts
CO2	Speak fluently and accurately in formal and informal communicative contexts
CO3	Respond to and express opinions effectively in Oral mediums of communication such as debates, Group Discussions, classroom discussions.

CO4	Differentiate among various types of listening and adopt various speaking strategies to present on different topics or excerpts.
CO5	Identify paralinguistic cues such as postures, gestures, facial expressions, and eye contact and speak effectively in debates and discussions.

Course Code & Title : HS3252 Professional English - II
Semester & Year of study : II & 2021-22

Course Outcomes	
CO1	Compare and contrast products and ideas in technical texts and write relevant texts.
CO2	Identify and report cause and effects in events, industrial processes through technical texts.
CO3	Analyse problems in order to arrive at feasible solutions and communicate them in the written format
CO4	Present their ideas and opinions in a planned and logical manner.
CO5	Draft effective resumes in the context of job search.
CO6	Detect paralinguistic cues such as postures, gestures, facial expressions, and eye contact.

Course Code & Title : MA3251 Statistics and Numerical Methods
Semester & Year of study : II & 2021-22

Course Outcomes	
CO1	Apply the concept of testing of hypothesis for small and large samples in real life problems.
CO2	Apply the basic concepts of classifications of design of experiments in the field of designing engineering problems.
CO3	Apply the numerical techniques for solving algebraic, transcendental equations, system of linear equations and eigenvalue problems.
CO4	Make use the numerical techniques of interpolation in various intervals and apply the numerical techniques of differentiation and integration for engineering problems
CO5	Apply the knowledge of various techniques and methods for solving first order ordinary differential equations with initial and boundary conditions in engineering problems.
CO6	Demonstrate the usage of MATLAB Software for solving algebraic and transcendental equations and Numerical Integration.

Course Code & Title : PH3254 Physics for Electronics Engineering
Semester & Year of study : II & 2021-22

Course Outcomes	
CO1	Elaborate the basics of crystallography and its importance for varied materials properties

CO2	Acquire knowledge on the electrical and magnetic properties of materials and their applications
CO3	Understand clearly of semiconductor physics and functioning of semiconductor devices
CO4	Understand the optical properties of materials and working principles of various optical devices
CO5	Express the importance of nanotechnology and nanodevices.

Course Code & Title : BE3254 Electrical and Instrumentation Engineering
Semester & Year of study : II & 2021-22

Course Outcomes	
CO1	Explain the working of transformer and its mathematical model
CO2	Elaborate the principles and characteristics of DC electrical machines
CO3	Analyze the performance of AC rotating machines.
CO4	Apply the concepts of measurements and instruments for real time applications.
CO5	Describe the basic power system concepts and methods
CO6	Measure the performance of DC machines

Course Code & Title : GE3251 Engineering Graphics
Semester & Year of study : II & 2021-22

Course Outcomes	
CO1	Construct the conic sectional curves, cycloidal and involute curves
CO2	Solve practical problems involving projection of lines
CO3	Construct the projection of simple solids and free hand sketches of multiple views from pictorial view of objects
CO4	Sketch the section of solids and Development of lateral surfaces of solids
CO5	Illustrate the isometric and perspective projections of simple solids
CO6	Sketch the real components of machine/ building/ circuit drawing based on the basic concept learned from the topics

Course Code & Title : EC3251 Circuit Analysis
Semester & Year of study : II & 2021-22

Course Outcomes

CO1	Apply the basic concepts of circuit analysis such as kirchoff’s law, mesh current and node voltage method for analysis of DC and AC circuits.
CO2	Apply suitable network theorems and analyze AC and DC circuits
CO3	Analyze steady state response of any R, L and C circuits
CO4	Analyze the transient response for any RC, RL and RLC circuits and frequency response of parallel and series resonance circuits.
CO5	Analyze the coupled circuits and network topologies
CO6	Construct a DC circuit simulation set up for implementing circuit laws and network theorems

Course Code & Title : GE3271 Engineering Practices Laboratory
Semester & Year of study : II & 2021-22

Course Outcomes	
CO1	Draw pipe line plan; lay and connect various pipe fittings used in common household plumbing work; Saw; plan; make joints in wood materials used in common household wood work.
CO2	Construct various electrical joints in common household electrical wire work.
CO3	Perform welding various joints in steel plates using arc welding work; Machine various simple processes like turning, drilling, tapping in parts; Assemble simple mechanical assembly of common household equipments; Make a tray out of metal sheet using sheet metal work.
CO4	Perform soldering and test simple electronic circuits; Assemble and test simple electronic components on PCB.
CO5	Demonstrate assemble and dismantle of LED TV and computer/ laptop components.
CO6	Model a circuit design using passive and active component in virtual lab.

Course Code & Title : EC3271 Circuits Analysis Laboratory
Semester & Year of study : II & 2021-22

Course Outcomes	
CO1	Verify Kirchhoff's Current Law and Kirchhoff's Voltage Law
CO2	Verify Thevenin and Norton Theorems
CO3	Verify Superposition, Maximum Power Transfer Theorems
CO4	Determine the resonant frequency of series and parallel RLC circuits
CO5	Determine the transient response of RL and RC circuits

CO6	Design and construct the RL and RC circuits and verify with simulation tool

Course Code & Title : GE3272 Communication Laboratory / Foreign Language
Semester & Year of study : II & 2021-22

Course Outcomes	
CO1	Speak effectively in group discussions held in a formal / semi-formal context
CO2	Discuss, analyse and present concepts and problems from various perspectives to arrive at suitable solutions
CO3	Write emails, letters and effective job applications
CO4	Write critical reports to convey data and information with clarity and precision
CO5	Give appropriate instructions and recommendations for safe execution of tasks
CO6	Identify paralinguistic cues such as postures, gestures, facial expressions, and eye contact and speak effectively in discussions and debates.

II Year

Course Code & Title : MA3355 Random Processes and Linear Algebra
Semester & Year of study : III & 2022-22

Course Outcomes	
CO1	Use the fundamental knowledge of the concepts of probability and standard distributions which can describe real life phenomenon.
CO2	Apply the basic concepts of one and two dimensional random variables in engineering applications.
CO3	Apply the concept of random processes in engineering disciplines.
CO4	Students will be able to explain the fundamental concepts of advanced algebra and their role in modern mathematics and applied contexts on the topic Vector Spaces.
CO5	Students will be able to demonstrate accurate and efficient use of advanced algebraic techniques and to demonstrate their mastery by solving non - trivial problems related to the concepts and by proving simple theorems about the statements proven by the text on the topic Linear Transformation and Inner Product Spaces.
CO6	Students will be able to collect data from a company and analyze using the concept of correlation and linear regression

Course Code & Title : CS3353 C Programming and Data Structures
Semester & Year of study : III & 2022-23

Course Outcomes	
CO1	Develop C programs for solving technical Application.

CO2	Apply advanced features for solving real world application.
CO3	Implement linear data structure operations using functions.
CO4	Use appropriate non–linear data structure operations for solving a given problem.
CO5	Apply appropriate sorting and searching algorithms for a given application.

Course Code & Title : EC3354 Signals and Systems
Semester & Year of study : III & 2022-23

Course Outcomes	
CO1	Analyze the classification of Signals and Systems
CO2	Apply Fourier Series, Laplace transform and Fourier transform in signal analysis.
CO3	Compute the output of Linear Time Invariant Continuous Time Systems using Laplace and Fourier Transforms.
CO4	Analyze discrete time signals using Z transform and Discrete Time Fourier Transform.
CO5	Compute the output of Linear Time Invariant Discrete Time Systems using Z transform and Discrete Time Fourier Transform.
CO6	Generate periodic signal and implement the Sampling process in time and frequency domain for various Nyquist rates

Course Code & Title : EC3353 Electronic Devices and Circuits
Semester & Year of study : III & 2022-23

Course Outcomes	
CO1	Illustrate the structure and working of basic electronic devices.
CO2	Analyze the frequency response of small signal amplifiers.
CO3	Analyze and design differential amplifier, single stage amplifier and multistage amplifier circuits
CO4	Analyze and design and analyze feedback amplifiers and oscillators circuits.
CO5	Analyze and design the various power amplifiers and DC/DC Converters
CO6	Design and construct the feedback amplifier circuit using SPICE tool

Course Code & Title : EC3351 Control Systems
Semester & Year of study : III & 2022-23

Course Outcomes

CO1	Compute transfer function modeling of different physical systems using various techniques.
CO2	Obtain the time response, identify the controllers and determine steady state error of control systems.
CO3	Illustrate the frequency response characteristics of the system using various plots and design various compensators.
CO4	Determine the stability of control systems using various methods.
CO5	Analyze the state space model using state variables and discuss the concepts of sampled data control system.
CO6	Construct the Time response and frequency response plot of control systems using MATLAB

Course Code & Title : EC3352 Digital Systems Design
Semester & Year of study : III & 2022-23

Course Outcomes	
CO1	Apply Boolean algebra and simplification procedures relevant to digital logic.
CO2	Design various combinational digital circuits using logic gates.
CO3	Analyse and design synchronous sequential circuits.
CO4	Analyse and design asynchronous sequential circuits
CO5	Design logic gates and use programmable devices
CO6	Design Digital transceiver/ALU and seven segment display decoder using logic gates.

Course Code & Title : EC3361 Electronic Devices and Circuits Laboratory
Semester & Year of study : III & 2022-23

Course Outcomes	
CO1	Examine the characteristics of PN Junction diode, Zener diode and Zener diode regulator.
CO2	Construct the Full wave rectifier with capacitive filter and sketch its response.
CO3	Investigate the input-output Characteristics of BJT and drain current and Transfer Characteristics of MOSFET.
CO4	Analyze the frequency response of CE, CB, CC ,CS and cascode amplifiers.
CO5	Design differential amplifier and calculate its CMRR.
CO6	Design and implement Class A Transformer Coupled Power Amplifier.

Course Code & Title : CS3362 C programming & Data Structures Laboratory **Semester & Year of study : III & 2022-23**	
Course Outcomes	
CO1	Develop C programs for given applications.
CO2	Implement Linear Data Structures using C
CO3	Implement Tree ADT for a given problem
CO4	Implement Searching and Sorting Techniques
CO5	Implement Various Hashing Techniques
Course Code & Title : EC3452 Electromagnetic Fields **Semester & Year of study : IV & 2022-23**	
Course Outcomes	
CO1	Apply the basic mathematical concepts of vector analysis
CO2	Describe the laws associated to static electric field and the properties of conductors and dielectrics.
CO3	Analyze the field potentials due to static magnetic fields and explain how materials affect electric and magnetic fields.
CO4	Analyze the relation between the fields under time varying situations and apply Maxwell's equations to electric and magnetic fields.
CO5	Explain electromagnetic wave propagation in lossy and in lossless media.
CO6	State the Significance of Maxwell's equation and explain their physical meaning
Course Code & Title : EC3401 Networks and Security **Semester & Year of study : IV & 2022-23**	
Course Outcomes	
CO1	Characterize the various types of Network Models, layers and its functions.
CO2	Illustrate the various network layer protocols and routing algorithms.
CO3	Examine the transport layer protocols and application layer paradigms.
CO4	Determine the network security architecture and algorithms.
CO5	Identify the hardware security attacks and its counter measures.

Course Code & Title : EC3451 Linear Integrated Circuits **Semester & Year of study : IV & 2022-23**	
Course Outcomes	
CO1	Explain about the basic building blocks of Linear Integrated Circuits.
CO2	Describe linear and non-linear applications of op-amps
CO3	Design applications using Analog multipliers and PLL.
CO4	Design ADC and DAC using op-amps
CO5	Generate waveforms using op-amp circuits and Analyze special function ICs
CO6	Design the filters using opamp
Course Code & Title : EC3492 Digital Signal Processing **Semester & Year of study : IV & 2022-23**	
Course Outcomes	
CO1	Apply DFT for the analysis of discrete signals & systems.
CO2	Design Infinite Impulse Response (IIR) filters based on the given specifications.
CO3	Design Finite Impulse Response (FIR) filters based on the given specifications.
CO4	Analyse the finite Word length effects in digital systems.
CO5	Perform multirate signal processing & characterize the functionalities and architecture of DSP processors.
Course Code & Title : EC3491 Communication Systems **Semester & Year of study : IV & 2022-23**	
Course Outcomes	
CO1	Analyze the various analog modulation techniques used for communication
CO2	Apply the concepts of Random Process, Sampling and Quantization to the design of communication systems
CO3	Apply the concepts of Pulse modulation and channel coding techniques
CO4	Develop applications using the various band pass signalling schemes.
CO5	Analyze the importance of demodulation techniques.

CO6	Simulate the various digital modulation techniques using MATLAB/Scilab/NI LabVIEW.

Course Code & Title : GE3451 Environmental Sciences and Sustainability
Semester & Year of study : IV & 2022-23

Course Outcomes	
CO1	Illustrate the importance of environment, biodiversity and its impact on human life.
CO2	Correlate the causes, effect and control measures of various types of pollution and its case studies.
CO3	Educate the availability of renewable energy resources and scientific concepts/principles behind them.
CO4	Articulate the concept of sustainability and management.
CO5	Gain knowledge on sustainability practices and socio-economical change.
CO6	Analyze the integrated themes of biodiversity, pollution control, waste management, energy resources, and sustainable approaches.

Course Code & Title : EC3461 Communication Systems Laboratory
Semester & Year of study : IV & 2022-23

Course Outcomes	
CO1	Design AM and FM for specific applications.
CO2	Compute the sampling frequency for digital modulation.
CO3	Implement Pulse digital modulation schemes
CO4	Implement PAM, PPM and PWM systems
CO5	Demonstrate their knowledge in base band signaling schemes through implementation of digital modulation schemes.
CO6	Apply various channel coding schemes & demonstrate their capabilities towards the improvement of the noise performance of Communication system.

Course Code & Title : EC3462 Linear Integrated Circuits Laboratory
Semester & Year of study : IV & 2022-23

Course Outcomes	
CO1	Analyze the basics of linear integrated circuits and various types of feedback amplifiers.
CO2	Design the oscillators, amplifiers, Wave shaping circuits and filters using operational amplifiers.
CO3	Analyze and implement the frequency multiplier using PLL.
CO4	Design D to A converters using Operational Amplifiers.

CO5	Analyze the performance of Oscillators, Amplifiers and Multivibrators using SPICE.
CO6	Build/test mini project using IC 741/555